

Healthcare Diagnosis Support

A Project Report

Submitted in the partial fulfillment of the requirements
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BACHELOR OF TECHNOLOGY

In

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Declaration

The Project Report entitled "**Healthcare Diagnosis Support**" is a record of Bonafide work of team members C. Keerthana– 2420030581, M. Varshitha – 2420030479, M. Vennela – 2420030218, K. Himavarsha – 2420030363, submitted in partial fulfilment for the award of B. Tech in Computer Science Engineering to K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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Certificate

This is to certify that the project-based report entitled " Healthcare Diagnosis Support " is a Bonafide work carried out and submitted by C. Keerthana– 2420030581, M. Varshitha – 2420030479, M. Vennela – 2420030218, K. Himavarsha – 2420030363, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science Engineering, at K L (Deemed to be University), during the academic year 2025-2026.

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ABSTRACT

Healthcare Diagnosis Support is a modern AI-powered web application designed to assist users in assessing the likelihood of diabetes using patient health metrics. The application provides a user-friendly interface where individuals can enter medical information such as glucose level, blood pressure, BMI, insulin level, age, and other relevant parameters. Based on these inputs, the system generates a diabetes risk prediction using a realistic mock machine learning model. The project is designed so that the current mock prediction logic can later be replaced with a real machine learning model developed using Python, Flask or FastAPI, and scikit-learn.

The application is developed using React, TypeScript, Tailwind CSS, React Router, React Hook Form, and Recharts to create a responsive and interactive user experience. It includes several modules such as user authentication, health dashboard, diabetes prediction, prediction history, AI health assistant, and user profile management. Interactive charts, progress indicators, and health statistics help users better understand their prediction results while maintaining a clean and accessible healthcare-themed interface. The primary objective of this project is to demonstrate how artificial intelligence can support early health risk assessment through an intuitive web application. Although the current implementation uses mock data, the application's modular architecture enables seamless integration with a real machine learning backend in future development. This project highlights the practical use of modern web technologies in healthcare applications while promoting awareness of diabetes risk and encouraging users to seek professional medical consultation when necessary.

INTRODUCTION

Diabetes is one of the most common chronic diseases worldwide and has become a major public health concern. It occurs when the body is unable to produce sufficient insulin or cannot effectively use the insulin it produces, resulting in high blood glucose levels. If left undiagnosed or untreated, diabetes can lead to severe health complications such as heart disease, kidney failure, nerve damage, vision loss, and stroke.

The Healthcare Diagnosis Support application is designed to provide an easy-to-use platform for assessing diabetes risk using commonly available health metrics. Users can enter medical parameters such as glucose level, blood pressure, BMI, insulin level, age, and diabetes pedigree function. The application processes these inputs and generates a prediction indicating whether the user is at low, moderate, or high risk of diabetes. In addition to the prediction, the system provides health insights, lifestyle recommendations, dietary suggestions, exercise advice, and consultation guidance. The project currently uses realistic mock prediction data but is structured to integrate a Python-based machine learning model in future versions.

Problem Description and Objectives

Many people remain unaware of their diabetes risk until the disease reaches an advanced stage. Existing healthcare systems often require clinical visits and laboratory tests before any assessment can be made, which may delay early detection. There is a need for an accessible and user-friendly application that can provide an initial diabetes risk assessment using readily available health information.

The proposed system addresses this problem by offering an AI-powered web application that enables users to enter their health metrics and receive an instant diabetes risk prediction along with personalized health recommendations. Although the prediction is simulated in the current version, the application's architecture is designed to support integration with a real machine learning model for accurate clinical predictions.

The main objectives of the project are:

- To develop a modern web application for diabetes risk prediction.
- To provide an intuitive and responsive user interface for entering health data.
- To validate user inputs using appropriate validation techniques.
- To generate diabetes risk predictions using realistic mock AI responses.
- To present prediction results using charts, progress indicators, and health statistics.
- To provide lifestyle, dietary, and exercise recommendations based on the prediction.
- To maintain prediction history for future reference.
- To design the application so that it can be integrated with a Python Flask or FastAPI machine learning backend.

LITERATURE SURVEY

Diabetes is one of the leading chronic diseases worldwide, affecting millions of people each year. Early identification of diabetes can significantly reduce the risk of severe complications by enabling timely medical intervention and lifestyle modifications. Over the years, researchers have developed several machine learning and artificial intelligence techniques to predict diabetes using patient health records and clinical data.

With the increasing availability of healthcare datasets and computational tools, AI-based disease prediction systems have become more accurate and accessible. These systems assist healthcare professionals by analyzing patient information and identifying patterns that may not be immediately noticeable through traditional methods. However, they are intended to support clinical decision-making rather than replace medical expertise.

Machine Learning Techniques Used

Common machine learning algorithms used for diabetes prediction include Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). Among these, Logistic Regression and Random Forest are widely used because they provide good prediction accuracy and are suitable for medical datasets. Among these methods, Random Forest and Logistic Regression are frequently used due to their balance of accuracy, efficiency, and interpretability.

Research Gap

Although numerous diabetes prediction systems have been proposed, many focus only on prediction accuracy and do not emphasize user experience or practical usability. Most applications simply display whether a person is diabetic or non-diabetic without explaining the prediction or suggesting preventive measures. These limitations indicate the need for a modern, user-friendly healthcare platform that combines prediction, visualization, health recommendations, and future-ready architecture.

Proposed System

The proposed Healthcare Diagnosis Support application addresses these limitations by providing a comprehensive web-based platform for diabetes risk assessment. The application accepts important health parameters from users and generates realistic AI-based prediction results. Along with the prediction, users receive confidence scores, health scores, risk levels, dietary recommendations, exercise suggestions, and lifestyle improvements.

The application also includes additional modules such as a health dashboard, AI Health Assistant, prediction history, and profile management, providing a more engaging and informative user experience.

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Specifications

The Healthcare Diagnosis Support application is designed as a web-based system and can be executed on any modern personal computer without requiring specialized hardware. Since the current version performs prediction using realistic mock data on the frontend, it does not require high computational resources. The modular architecture also allows future integration with a Python-based machine learning backend without significant hardware changes.

Processor

A system with an Intel Core i3 processor or equivalent is sufficient to run the application efficiently. For development and future backend integration involving machine learning models, an Intel Core i5 processor or higher is recommended to ensure smooth performance.

RAM

A minimum of 4 GB RAM is required to run the development environment and the web application. 8 GB RAM or higher is recommended for faster development, multitasking, and running both the frontend and backend services simultaneously.

Storage

Approximately 2 GB of free disk space is recommended for installing project dependencies, development tools, and source code. Additional storage may be required when integrating machine learning models and datasets in future versions.

Network Connectivity

A stable internet connection is required for installing project dependencies using npm, accessing development resources, and deploying the application. Once installed, the frontend application can run locally without continuous internet access.

Operating System

The application is compatible with Windows, Linux, and macOS. Since it is developed using cross-platform web technologies, it can be executed on any operating system that supports Node.js and modern web browsers.

Software Requirements

The application is developed using modern web technologies that provide a responsive, scalable, and user-friendly interface. The software components used in the project are described below.

Programming Languages

The frontend is implemented using TypeScript with React, while future backend integration is planned using Python with Flask or FastAPI for machine learning-based diabetes prediction.

Libraries and Frameworks

- React – Used for building reusable user interface components and managing application views.
- TypeScript – Provides type safety and improves code reliability.
- Tailwind CSS – Used for responsive styling and modern user interface design.
- React Router – Handles navigation between application pages.
- Recharts – Used to display health statistics and prediction analytics through interactive charts.

Development Environment

The project is developed using Visual Studio Code as the primary code editor. Node.js and npm are used for package management and running the development server. Git and GitHub are used for version control and project collaboration.

Future Backend Support

The application architecture is designed to support integration with a Python Flask or FastAPI backend. A machine learning model developed using scikit-learn can replace the current mock prediction logic, enabling real-time diabetes prediction based on patient health metrics.

With the above hardware and software configuration, the Healthcare Diagnosis Support application can be developed, tested, and executed efficiently on any modern personal computer. The use of lightweight frontend technologies ensures smooth performance, while the modular architecture allows seamless integration of advanced machine learning models in future development.

IMPLEMENTATION

The Healthcare Diagnosis Support application is developed as a modern web-based platform that assists users in assessing their diabetes risk using health-related parameters. The application follows a modular architecture, making it easy to maintain and extend. Each module performs a specific function while working together to provide a seamless user experience.

Landing Page

The landing page serves as the entry point of the application. It introduces the purpose of the Healthcare Diagnosis Support system and provides an overview of its features. The page includes a hero section, project description, diabetes awareness information, feature highlights, and a call-to-action button that directs users to the authentication page. The responsive design ensures accessibility across desktop and mobile devices.

Authentication Module

The authentication module allows users to register, log in, and recover their passwords. Since the current implementation is frontend-based, mock authentication is used to simulate user login and account management. Protected routes ensure that only authenticated users can access the dashboard and prediction features.

Dashboard

After successful login, users are redirected to the dashboard. This module provides an overview of the user's health activities through summary cards displaying the total number of predictions, recent prediction history, health statistics, and risk distribution. Interactive charts help users understand their health trends, while quick action buttons provide easy navigation to other modules.

Diabetes Prediction Module

The diabetes prediction module is the core component of the application. Users enter health parameters such as pregnancies, glucose level, blood pressure, skin thickness, insulin level, BMI, diabetes pedigree function, and age through a validated form. The entered values are processed using realistic mock prediction logic to generate the likelihood of diabetes.

The prediction result includes the diabetes status, confidence percentage, health score, risk level, and personalized recommendations related to diet, exercise, lifestyle improvements, and medical consultation. A disclaimer is displayed to inform users that the prediction is intended only for educational purposes and should not replace professional medical advice.

Prediction History

This module stores all previous prediction results using mock data. Users can search, filter, and review earlier predictions, allowing them to monitor changes in their health assessments over time. The module also provides an option to export prediction reports in PDF format.

AI Health Assistant

The AI Health Assistant provides a chatbot interface where users can ask general health-related questions. The chatbot responds with predefined mock responses and offers useful health tips, diabetes awareness information, and answers to frequently asked questions. Chat history is maintained during the user session to improve the interaction experience.

Profile Module

The profile module allows users to view and update their personal information, health metrics, and prediction statistics. It also provides application settings and user preferences, creating a personalized experience for each user.

Future Backend Integration

The application is designed using a modular architecture that supports seamless integration with a Python Flask or FastAPI backend. In future development, the current mock prediction logic can be replaced with a trained scikit-learn machine learning model. The frontend will send user health metrics to the backend through REST APIs, receive prediction results, and display them in the existing user interface without requiring major architectural changes.

EXPERIMENTATION AND CODE

The Healthcare Diagnosis Support application was tested to ensure that all modules function correctly and provide a smooth user experience. Each feature was verified individually to check navigation, form validation, prediction generation, dashboard updates, and responsiveness across different screen sizes.

The diabetes prediction form was tested using various valid and invalid input values. Input validation was implemented using React Hook Form and Zod to ensure that users enter appropriate health metrics before a prediction is generated. Invalid inputs display meaningful error messages, preventing incorrect data from being processed.

The prediction system currently uses realistic mock data to simulate the output of a machine learning model. Based on the user inputs, the application generates a prediction result that includes the diabetes status, confidence percentage, health score, risk level, and personalized health recommendations. This mock implementation can later be replaced by a Flask or FastAPI backend connected to a scikit-learn model.

The dashboard was tested to verify that health statistics, prediction history, and charts update correctly after each prediction. The AI Health Assistant was also tested using predefined health-related questions to ensure consistent mock responses and proper chat history management.

The following code snippet illustrates the prediction function used in the current implementation.

```
const handlePrediction = (data: PredictionInput) => {  
  const isDiabetic = data.glucose > 140 && data.bmi > 30;  
  const result = {  
    prediction: isDiabetic ? "Diabetic" : "Non-Diabetic",  
    confidence: isDiabetic ? 91 : 87,  
    riskLevel: isDiabetic ? "High" : "Low",  
    healthScore: isDiabetic ? 42 : 88  
  };  
  setPrediction(result);  
};
```

The application was also tested for responsiveness on desktop and mobile devices to ensure a consistent user experience. Navigation between pages, protected routes, and mock authentication were verified to function correctly without runtime errors.

Overall, the experimental results demonstrate that the application performs as expected, providing an intuitive interface for diabetes risk assessment while maintaining a modular structure that supports future integration with a real machine learning backend.

RESULTS

The Healthcare Diagnosis Support application was successfully developed and tested using realistic mock data. All modules functioned as expected, providing a responsive and user-friendly interface for diabetes risk assessment. The application allows users to register, log in, enter health metrics, and receive an instant prediction along with personalized health recommendations.

The dashboard successfully displays prediction statistics, recent prediction history, and health analytics through interactive charts. The diabetes prediction module validates user inputs before generating a prediction result that includes the diabetes status, confidence percentage, health score, and risk level. Additional recommendations such as diet suggestions, exercise plans, lifestyle improvements, and doctor consultation advice are presented to help users better understand their health condition.

The AI Health Assistant responds to predefined health-related queries and provides useful health tips, while the prediction history module stores previous predictions for future reference. The application is fully responsive and performs efficiently across different screen sizes and modern web browsers.

The project also demonstrates a modular architecture that can be easily integrated with a Python Flask or FastAPI backend and a scikit-learn machine learning model, enabling real-time diabetes prediction in future development.

MediPredict | Diabetes Prediction

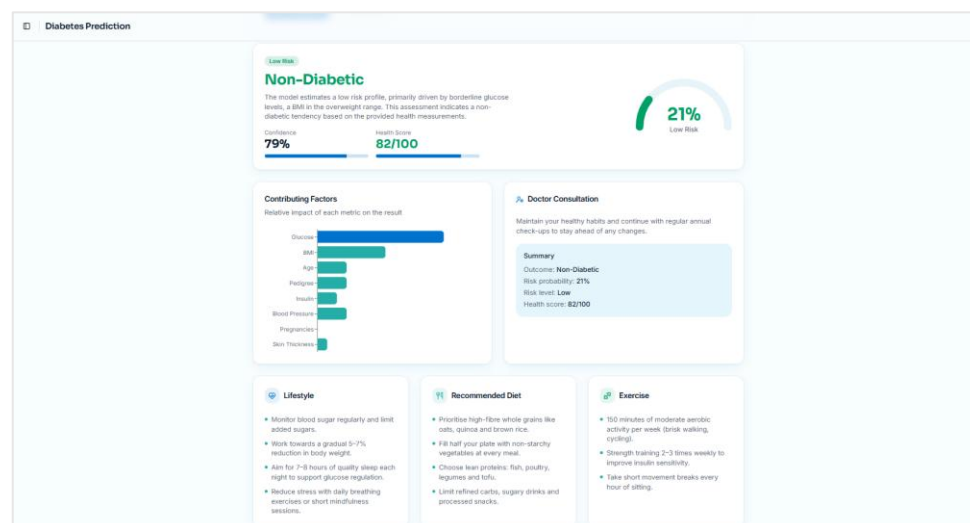
Menu: Dashboard, New Prediction, History, AI Assistant, Profile

Patient Health Metrics
Enter the values below to generate an AI diabetes risk assessment.

Pregnancies 0 Number of times pregnant	Glucose Level 120 Plasma glucose (mg/dL)
Blood Pressure 72 Diastolic (mm Hg)	Skin Thickness 25 Triceps fold (mm)
Insulin 100 2-hour serum (mu U/ml)	BMI 27.5 Body mass index
Diabetes Pedigree Function 0.45 Family history score	Age 35 Age in years

Predict **Reset**

Keerthana
c.keerthana08@gmail.com
Sign out



CONCLUSION

The Healthcare Diagnosis Support application successfully demonstrates how modern web technologies can be used to develop an AI-assisted healthcare platform for diabetes risk assessment. The system provides an intuitive interface that enables users to enter health metrics and receive a prediction along with confidence scores, health insights, and personalized recommendations. Features such as user authentication, dashboard analytics, prediction history, and an AI Health Assistant enhance the overall user experience and make the application more informative and interactive.

Although the current implementation uses realistic mock prediction logic, the application is designed with a modular architecture that allows seamless integration with a Python Flask or FastAPI backend and a scikit-learn machine learning model. Future enhancements may include real-time disease prediction, secure database integration, advanced AI chat support, and additional disease prediction modules. Overall, the project demonstrates the potential of AI-powered web applications in supporting early health awareness and promoting preventive healthcare.

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